Higher Topos Theory - Self Study

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Contents

Abstract

I am currently reading Higher Topos Theory to understand the theory of quasi-categories and basic ∞-category theory. My goal is to understand the generalizations of basic 1-categorical concepts to the ∞-categorical setting, such as (co)limits, adjunctions, and Yoneda's lemma. I'm also interested in more advanced topics in higher category theory, most of which require a basic understanding of the theory of ∞-categories as a prerequisite.

I began reading this text on August 8th, 2025, at 11:06 PM, as an independent project. I am hoping to be able to find a professor who is willing to support my readings, but I am unsure that will happen. Nevertheless, I intend to gain a working understanding of ∞-categories over the next few months, and a more advanced understanding come the spring semester. Since this text is quite advanced and is not meant as a textbook, but rather as a reference, I will be taking extremely detailed and extensive notes on the text. I will also be referencing supplementary resources throughout my reading, so I have decided it worthwhile to maintain a bibliography for these notes. Any notes not credited to a bibliography source are either from [Lur09], or directly from me.

Chapter 1

An Overview of Higher Category Theory

The first chapter introduces quasi-categories as models for ∞-categories, and then develops ∞-categorical analogues of standard 1-categorical notions.

1.1 Foundations of Higher Category Theory

Terminology 1.1.0.1. Throuhgout this paper, the term *space* is short for a topological space, and almost always a compactly generated topological space. The *homotopy type* of a topological space is a choice of a weak homotopy equivalent topological space, or equivalently the collection of weak homotopy equivalent spaces.

1.1.1 Goals and Obstacles

The theory of ∞-categories finds natural motivation in the fundamental n-groupoids of a topological space X. Define $\Pi_n(X)$ to take points $x \in X$ as objects, paths from x to y as 1-morphisms, homotopies of paths as 2-morphisms, and so on until n. Since vertical composition of homotopies is only associative up to higher homotopy, we are forced to take n-morphisms to be *homotopy classes*, rather than simply homotopies. Of course $\Pi_n(X)$ is a groupoid because all paths and homotopies admit inverses.

One might hope that we can fix this problem by considering an ∞-groupoid, or an $(\infty, 0)$-category, which would not take homotopies as n-morphisms for all $n \geq 2$. Interestingly, the ∞-groupoid $\Pi_\infty(X)$ of a space X encodes the homotopy type of X, meaning the study of ∞-groupoids simplifies to the study of spaces in classical homotopy theory, and we have many ways of describing homotopy types of spaces. It is also a commonly accepted principle of higher category theory that all $(\infty, 0)$-categories are of the form $\Pi_\infty(X)$ for some space X. This principle is profound, since it gives us a place to begin our work in higher category theory.

In general, an n-category should be a category enriched over $(n-1)$-categories. However, for finite n, this process produces the notion of a *strict* n-category, and n-category theory is more naturally viewed via the theory of *weak* n-categories. These issues make the theory of n-categories fairly difficult to work with, but they vanish when one passes to ∞-categories. An (∞, n)-category can indeed be viewed as a category enriched over $(\infty, n-1)$-categories. This discussion suggests a definition (or more precisely, a blueprint) for ∞-categories. An ∞-category $\mathcal{C}$ should consist of objects $\mathrm{Obj}(\mathcal{C})$, and for each $x, y \in \mathcal{C}$, a ∞-groupoid $\mathrm{Map}_{\mathcal{C}}(x, y)$ with an associative composition law. The theory of ∞-categories does not care whether this law be strictly associative or only associative up to coherent homotopy, since any ∞-category with coherently associative multiplication is equivalent to one with strictly associative multiplication.

Definition 1.1.1.1. A *topological category* is a category enriched over $\mathcal{CG}$, the category of compactly generated spaces. The category of topological spaces will be denoted by $\mathcal{C}\mathrm{at}_{\mathrm{top}}$.

We recall from our knowledge of enriched category theory that this implies a topological category $\mathcal{C}$ has topological spaces $\mathrm{Map}_{\mathcal{C}}(x, y)$ for each pair $x, y \in \mathcal{C}$, with an associative composotion law

$$\mathrm{Map}_{\mathcal{C}}(y, z) \times \mathrm{Map}_{\mathcal{C}}(x, y) \to \mathrm{Map}(x, z),$$

where the product is computed in $\mathcal{CG}$.

Remark 1.1.1.2. As is often the case, working in $\mathcal{CG}$ is done largely as a technical point which can be ignored.

1.1.2 ∞-Categories

Definition 1.1.1.1 is the most obvious definition of an ∞-category, but it is also one of the hardest to actually work with. Our use of topological categories in this book will be more as a "benchmark" of sorts. Any model for ∞-categories which is equivalent to that of topological categories must necessarily be a good one. Many better models exist, and in this text we choose to use the theory of *weak Kan complexes*. These were first introduced by Boardman and Vogt in [BV73], and were more recently and more extensively studied by André Joyal in [Joy08a, Joy08b], who instead called them *quasi-categories*, which the modern literature now uses. However, Lurie's book almost exclusively uses quasi-categories as our model for ∞-categories, so for the purposes of these notes we will use the terms "quasi-category" and "∞-category" more or less interchangeably.

Quasi-categories are built atop the theory of *simplicial sets*, and any readers of HTT (or these notes) should have at least an elementary understanding of the subject. I recommend my own exposition [Tal25], which builds the abstract theory of simplicial sets without forsaking geometric intuition, but also without dwelling excessively on the topological impliciations of the theory. We also recommend [Rie11, Fri23]. The former should prepare the reader for much of the theory in this paper, the rest of which can be picked up along the way, although we feel Riehl strays too far from the geometric intuition of simplices. Conversely, Friedman's introduction is well-liked for the geometric intuition it provides, but we feel it focuses too much on topological aspects of the theory to be the best preparation for the material covered in HTT.

The theory of simplicial sets is of importance due to the homotopic aspects of the theory. Recall from our discussion in section 1.1.1 that the ∞-groupoid $\Pi_\infty(X)$ of a space X captures the homotopy type of X. We also know from [Qui67] that since $|-| \dashv \mathrm{Sing}$ is a Quillen equivalence, where Sing is the singular complex functor. This means in particular that the unit and counit of the adjunction are natural weak homotopy equivalences, meaning spaces are determined by their singular sets up to homotopy type. Thus, if we are only interested in the study of spaces up to weak homotopy equivalence, we might as well work in the category of simplicial sets. We might also think that ∞-groupoids should be able to be modeled by singular sets, since they both encode the homotopy type of a space. Indeed this is possible, but we prefer to work slightly more generally. Singular sets satsify the following important property.

Definition 1.1.2.1. Let K be a simplicial set. We say K is an *Kan complex* if, for all $n \geq 0$ and for all $0 \leq i \leq n$, any diagram of solid arrows

$$\begin{tikzcd} \Lambda^n_i \arrow[r] \arrow[d, hook] & X \\ \Delta^n \arrow[ur, dotted] & \end{tikzcd}$$

admits a (not necessarily unique) dotted arrow as indicated, rendering the diagram commutative.

Recall here that $\Lambda^n_i \subseteq \Delta^n$ represents the ith horn of the standard n-simplex; we defer the reader to [Tal25, §3.4] for a rigorous definition.

Remark 1.1.2.2. The singular complex of a topological space is a Kan complex. Showing this first involves showing that $|\Lambda^n_i|$ is a retract of $|\Delta^n|$ in $\mathcal{T}\mathrm{op}$. There also exist combinatorial approaches to extracting homotopy groups from any Kan complex K, which agree with the homotopy groups on $|K|$. Thus, Kan complexes behave like the homotopy type of a space.

Simplicial sets, and in particular Kan complexes, thus should form a model for ∞-groupoids, which we note are an important special case of ∞-categories. Indeed, Kan complexes are a common model for ∞-groupoids, which we will explore later. However, another special case of ∞-categories that our definition must reflect is the notion of a basic 1-category. If an ∞-category $\mathcal{C}$ has no nontrivial n-morphisms for $n > 1$, then clearly $\mathcal{C}$ can be regarded as a 1-category. Luckily, simplicial sets are also closely related to category theory. Recall the nerve $N(\mathcal{C})$ of a category is a simplicial set whose n-simplices are functors $[n] \to \mathcal{C}$, which carry the data of an n-tuple of composable morphisms.

Remark 1.1.2.3. The nerve encodes the full information of a category, and we can in fact recover the initial category $\mathcal{C}$ from the nerve $N(\mathcal{C})$ with the following list of observations.

1. The 0-simplices $N(\mathcal{C})_0$ are the objects, so we define $\mathrm{Obj}(\mathcal{C}) = N(\mathcal{C})_0$.
2. The 1-simplices $N(\mathcal{C})_1$ are simply morphisms, so we define $\mathrm{Mor}(\mathcal{C}) = N(\mathcal{C})_1$.
3. Given an object $x \in \mathcal{C}$, the degeneracy $s_0 : N(\mathcal{C})_0 \to N(\mathcal{C})_1$ maps to the identity $x \mapsto 1_x$. Thus, the identity function $x \mapsto 1_x$ is simply the degeneracy $s_0 = 1_\bullet$.
4. Given a morphism $f \in N(\mathcal{C})_1$, the 0th face $d_0 : N(\mathcal{C})_1 \to N(\mathcal{C})_0$ maps to the codomain $f \mapsto \operatorname{cod} f$, and similarly the first face $d_1 : N(\mathcal{C})_1 \to N(\mathcal{C})_0$ maps to the domain $f \mapsto \operatorname{dom} f$. We may thus define $\operatorname{dom}, \operatorname{cod} : \mathrm{Mor}(\mathcal{C}) \to \mathrm{Obj}(\mathcal{C})$ as the faces $\operatorname{dom} = d_1$ and $\operatorname{cod} = d_0$.
5. A composable pair (g, f) is a 2-simplex $(g, f) \in N(\mathcal{C})_2$. The 1st face $d_1 : N(\mathcal{C})_2 \to N(\mathcal{C})_1$ maps composable pairs to their composite $(g, f) \mapsto gf$. Thus, composition is simply the 1st face $d_1 = \circ$.
6. Composites of larger tuples of maps arise from higher degeneracies, and associativity follows from the simplicial identities. The unit law for identities under composition also follows from simplicial identities.

A more explicit description of remark 1.1.2.3 (5,6) will be helpful later. A morphism from x to y is defined as a 1-simplex $f \in N(\mathcal{C})_1$ with $d_1(f) = x$ and $d_0(f) = y$. Given two morphisms $x \xrightarrow{f} y \xrightarrow{g} z$, the composite $g \circ f$ can be characterized by noting there exists a 2-simplex $(g, f) = \sigma \in N(\mathcal{C})_2$ such that

1. $d_0(\sigma) = g$;
2. $d_1(\sigma) = gf$;

3. $d_2(\sigma) = f$.

We say σ *witnesses gf as a composite of g and f.*

There exists a characterization of simplicial sets that arise as nerves, which is more straightforward than the characterization of Kan complexes.

Notation 1.1.2.4. Let J be a linearly ordered set. Denote by Δ^J the representable functor $\mathrm{Hom}(-, J)$ in the *category of linearly ordered sets*, restricted along the objects of $\mathbf{\Delta}$. Clearly, $\Delta^{[n]} = \Delta^n$, so this is simply a mild generalization of notation we already had.

Proposition 1.1.2.5. *Let K be a simplicial set. The following conditions are equivalent.*

1. *There exists a small category $\mathcal{C}$ and an isomorphism $K \cong N(\mathcal{C})$.*
2. *For each $0 < i < n$, and each diagram of solid arrows*

$$\begin{tikzcd} \Lambda^n_i \arrow[r] \arrow[d, hook] & K \\ \Delta^n \arrow[ur, dashed] & \end{tikzcd}$$

there is a unique *dashed arrow as indicated, rendering the diagram commutative.*

Proof. We mostly follow Lurie's proof in [Lur09, Prop. 1.1.2.2], but we take a different route for showing agreement for (1), and Lurie assumes a slightly more advanced understanding of simplicial sets than I currently have, so at some points in this proof we may prove extra facts. For example, we start by proving that for $0 < i < n$, the vertices of the ith horn are $(\Lambda^n_i)_0 = [n]$. Note that for i to exist, we must have $n \geq 2$. For $n = 2$, there exist at two coface maps $d^j : \Delta^1 \to \Delta^2$ with $j \neq i$. 0-simplices of Δ^1 are arrows $[0] \to [1]$, which correspond to elements of $[1]$. More precisely, arrows are uniquely determined by their image. Thus, 0-simplices in the face $\partial_j \Delta^n_0 = \mathrm{im}\, d^j_0$ will also be uniquely determined by their image, so it suffices to show each $x \in [2]$ lands in the image of some coface d^j with $j \neq i$. This is immediate: since d^j only skips j, and since another coface d^k, $k \neq i$ exists which doesn't skip j, every element shows up. Inducting on this same logic proves the statement for $n \geq 2$, and vacuously for all n.

Now we show for $0 < k \leq n$ and $n \geq 2$, there is a simplicial subset $\Delta^{\{k-1,k\}} \subseteq \Lambda^n_i$. Recall that m-simplices f in the horn are composites $[m] \xrightarrow{f} [n-1] \xrightarrow{d^{j \neq i}} [n]$. Let $g : [m] \to \{k-1, k\} \subseteq [n]$ be an m-simplex in $\Delta^{k-1,k}$. If $k < n$, then $k, k-1 \in [n-1]$. Write g' for the map with codomain viewed as a subset $\{k-1, k\} \subseteq [n-1]$. Since $i \neq n$, we have that $\mathrm{im}\, d^n \subseteq \Lambda^n_i$. Thus, $d^n g' = g \in \mathrm{im}\, d^n_m$, and thus is an m-simplex in the horn. If $k = n$, then since $n \geq 2$, we have $k - 1 > 0$. Since $i \neq 0$, we do the same argument.

If $n < 2$ then the statement is true vacuously, so let $n \geq 2$. Let K be the nerve of a small category $\mathcal{C}$. Let $f_0 : \Lambda^n_i \to K$ with $0 < i < n$, and let $x_k \in \mathcal{C}$ denote the image of the vertex $k \in (\Lambda^n_i)_0 = [n]$ under f_0. Choose some $0 < k \leq n$, and consider the restriction $f_0|\Delta^{\{k-1,k\}}$. Note that the 1-simplices of $\Delta^{\{k-1,k\}}$ are

$$\Delta^{\{k-1,k\}}_1 = \left\{ (0,1) \mapsto k, (0,1) \mapsto k-1, \begin{matrix} 0 \mapsto k-1 \\ 1 \mapsto k \end{matrix} \right\}.$$

The first two are easily seen to be the degenerates $s_0(k), s_0(k-1)$, respectively. Since f_0 is a morphism of simplicial sets, $f_0 s_0(k) = s_0 f_0(k) = s_0(x_k) = 1_{x_k}$. This is uninteresting, so let g_k denote the unique nondegenerate simplex. Note that the image under f is a 1-simplex in $N(\mathcal{C})$; that is a morphism $g_k : c \to d$. We want to show that with notation as above, $c = x_{k-1}$ and $d = x_k$. Indeed, note that for 1-simplices in the nerve, the first face is simply *domain* and the 0th face is *codomain*. Since f_0 commutes with faces, we have

$$\operatorname{dom} g_k = d_1 f_0(g_k) = f_0 d_1(g_k) = f_0(g_k \circ d^1) = f_0(0 \mapsto 0 \mapsto k-1) = x_{k-1},$$

since $0 \mapsto 0 \mapsto k-1 = g_k \circ d^1$ is a 0-simplex, which is uniquely determined (and identified with) its image. The same follows for domain.

We thus have for each $0 < k \leq n$, there is an arrow $g_k : x_{k-1} \to x_k$, allowing us to form a composable chain

$$x_0 \xrightarrow{g_1} x_1 \xrightarrow{g_2} \cdots \xrightarrow{g_n} x_n.$$

This is an n-simplex in Δ^n. We can similarly construct m-simplices for $m \leq n$ by requiring $k \leq m$. We claim that this relation allows us to define a mapping $f : \Delta^n \to K$.

To see how this follows, we invoke the Yoneda lemma to recall that there is an isomorphism $\mathcal{S}\mathrm{et}^{\mathbf{\Delta}^{\mathrm{op}}}(\Delta^n, K) \cong K_n$ given by evaluating the nth component of natural transformations at the identity. In particular, Yoneda says a morphism $f : \Delta^n \to K$ is determined by $f_n(1_{[n]})$. We choose $f_n(1_{[n]})$ to be the n-simplex $(g_1, \ldots, g_n)$ of $N(\mathcal{C})$ considered above. The remainder of the proof will focus on showing this definition indeed agrees with Λ^n_i. For now, we note that f is manifestly unique, since we obtained a unique composite constructed in this way, so any other map $\Delta^n \to K$ which restricts along the horn to f_0 must necessarily correspond to the same chain of morphisms.

We now begin to show $f|\Lambda^n_i = f_0$. First, we show they agree on cofaces. Let $d^j : [n-1] \to [n]$ be an $(n-1)$-simplex in Λ^n_i. Then since f is a morphism of simplicial sets and commutes with faces, $d^j = 1_{[n]} d^j = d_j(1_{[n]})$ implies that $f(d^j) = d_j f(1_{[n]})$. For simplicity, we take d^j to be an inner face $0 < j < n$, and simply note the argument for outer faces is the same. From our knowledge of the nerve, we have that

$$f(d^j) = f d_j(1_{[n]}) = d_j f(1_{[n]}) = d_j \left(x_0 \xrightarrow{g_1} \cdots \xrightarrow{g_n} x_n \right)$$

$$= \ x_0 \xrightarrow{g_1} \cdots \xrightarrow{g_{j-1}} x_{j-1} \xrightarrow{g_{j+1} g_j} x_{j+1} \xrightarrow{g_{j+1}} \cdots \xrightarrow{g_n} x_n$$

It suffices to show $f_0(d^j)$ agrees with this composite. One can do this by showing each object and morphism align with that of $f(d^j)$, which involves an induction argument that is extremely lengthy due to needing to show many cases, but is in general very simple. We trust the reader to be able to do this argument, and we also trust that the reader knows I can do the argument too and just don't want to, so we prefer not to spend 2 pages on it and instead continue with the proof.

Now, we show this agreement on cofaces implies f and f_0 agree in general. We note that every m-simplex $h \in \Delta^n_m$ is a morphism $h : [m] \to [n]$ in $\mathbf{\Delta}$, which admits a factorization into cofaces and codegeneracies $h = d^{j_1} \circ \cdots d^{j_\ell} \circ s^{p_1} \circ \cdots \circ s^{p_q}$. Since faces and degeneracies in Δ^n are given by precomposing by the corresponding cofaces and codegeneracies in $\mathbf{\Delta}$, we have

$$h = d^{j_1} \circ \cdots \circ s^{p_q} = 1_{[n]} \circ d^{j_1} \circ \cdots \circ s^{p_q} = (s_{p_q} \circ \cdots \circ d_{j_1})(1_{[n]}).$$

Since f is a morphism of simplicial sets, we have $f s_j = s_j f$, and likewise for faces. Induction gives

$$f(h) = (f \circ s_{p_q} \circ \cdots \circ d_{j_1})(1_{[n]}) = (s_{p_q} \circ \cdots \circ d_{j_1})(f(1_{[n]})) = (s_{p_q} \circ \cdots \circ d_{j_2})(f(d_{j_1}(1_{[n]}))).$$

Now we want to show f agrees with f_0 on m-simplices of Λ^n_i. Since $h \in (\Lambda^n_i)_m$, there is some m-simplex h' such that $h = d^j h'$ for $j \neq i$, so by uniqueness we have that $d^{j-1} \neq d^i$ in the epi-mono factorization of h. Since f and f_0 agree on cofaces,

$$\begin{aligned} f(h) &= (s_{p_q} \circ \cdots \circ d_{j_2})(f(d_{j_1}(1_{[n]}))) \\ &= (s_{p_q} \circ \cdots \circ d_{j_2})(f(d_{j_1})) \\ &= (s_{p_q} \circ \cdots \circ d_{j_2})(f_0(d_{j_1})) \\ &= (s_{p_q} \circ \cdots \circ d_{j_2})(f_0(d_{j_1}(1_{[n]}))) \\ &\overset{!}{=} f_0(h), \end{aligned}$$

where the equality marked by ! follows since f_0 is a morphism of simplicial sets and commutes with faces and degeneracies. Thus, the statement ($1 \implies 2$) is proven.

Now for the converse. Let K satisfy (2), and define a category $\mathcal{C}$ as follows.

1. Objects are vertices: $\mathrm{Obj}(\mathcal{C}) = K_0$;
2. Morphisms are edges: $\mathrm{Mor}(\mathcal{C}) = K_1$. Domain and codomain are faces: $\mathrm{dom} = d_1$ and $\mathrm{cod} = d_0$;
3. Identity $1_\bullet : \mathrm{Obj}(\mathcal{C}) \to \mathrm{Mor}(\mathcal{C})$ is the degeneracy $1_\bullet = s_0$;
4. Let $f : x \to y$ and $g : y \to z$ in $\mathcal{C}$. We want to define a morphism $\sigma_0 : \Lambda^2_1 \to K$. Define the action on 0-simplices by $0 \mapsto x$, $1 \mapsto y$, and $2 \mapsto z$. Denote by g_k the nondegenerate simplex in $\Delta^{k-1,k}$ for $0 < k \leq 2$. Then define the map as $g_1 \mapsto f$ and $g_2 \mapsto g$. We will show later that this data determines a morphism $\sigma_0 : \Lambda^2_1 \to K$. By the horn filling condition, this extends to a map $\sigma : \Delta^2 \to K$. Let $d^1 : [1] \to [2]$ be the unique 1-simplex of Δ^2 with $d^1(0) = 0$ and $d^1(1) = 2$. Then define $g \circ f$ as $\sigma(d^1)$.

Of course, we need to show the data in (4) defines a morphism $\sigma_0 : \Lambda^2_1 \to K$. Recall that $\Lambda^2_1 = \partial_0\Delta^2 \cup \partial_2\Delta^2$. By universality of union, if we can define maps $\partial_0\Delta^2 \to K$ for $i = 0, 2$, such that these maps agree on $\partial_0\Delta^2 \cap \partial_2\Delta^2$, then they assemble into a map Λ^2_1. We may thus construct a map in this way using only the data of (4). Recall that $\partial_i\Delta^n = \mathrm{im}(d^i : \Delta^{n-1} \to \Delta^n)$, and in particular $d^i : \Delta^{n-1} \to \partial_i\Delta^n$ is an isomorphism by uniqueness of the epi-mono factorization in $\boldsymbol{\Delta}$, with a well-defined inverse $d^i \circ h \mapsto h$. Thus, if $\tau_0 : \Delta^{n-1} \to K$ is a morphism of simplicial sets, then there is a morphism of simplicial sets $\tau : \partial_i\Delta^n \to K$ given by $d^i \circ h \mapsto h \mapsto f_0(h)$, since this is simply composition of the inverse of the isomorphism with the map f_0. In particular, maps $\Delta^1 \to K$ induce maps $\partial_i\Delta^2 \to K$.

By the Yoneda lemma, maps $\tau_0' : \Delta^1 \to K$ are specified by the image $\tau_0'(1_{[1]})$. Thus, the map $\tau_0 : \partial_i\Delta^2 \to K$ is fully specified by the image $\tau_0'(1_{[1]})$ in the aforementioned manner. Define

$$\alpha_0 : \Delta^1 \to K, \qquad 1_{[1]} \mapsto f,$$

$$\beta_0 : \Delta^1 \to K, \qquad 1_{[1]} \mapsto g.$$

These of course assemble into maps $\alpha : \partial_2\Delta^2 \to K$ and $\beta : \partial_0\Delta^2 \to K$. To show these maps agree on the intersection, it suffices to show they agree on the vertex $1 \in \partial_{0,2}\Delta^2_0$. Indeed, $1 \in \partial_0\Delta^2_0$ is simply $d^0(0)$, and $1 \in \partial_2\Delta^2_0$ is simply $d^2(1)$. Thus, they are mapped

$$\alpha : 1 = d^2(1) \mapsto 1 \mapsto \alpha_0(1), \qquad \beta : 1 = d^0(0) \mapsto 0 \mapsto \beta_0(0).$$

We simply need to show these agree. Since β_0 and α_0 are morphisms of simplicial sets, they necessarily commute with faces, meaning the diagrams

$$\begin{tikzcd} \Delta^1_1 \arrow[r, "\alpha_0"] \arrow[d, "d_0"'] & K_1 \arrow[d, "d_0"] \\ \Delta^1_0 \arrow[r, "\alpha_0"'] & K_0 \end{tikzcd} \qquad \begin{tikzcd} \Delta^1_1 \arrow[r, "\beta_0"] \arrow[d, "d_1"'] & K_1 \arrow[d, "d_1"] \\ \Delta^1_0 \arrow[r, "\beta_0"'] & K_0 \end{tikzcd}$$

commute. Recall that $\alpha_0(1_{[1]}) = f$ and $\beta_0(1_{[1]}) = g$. Following the identity around both squares yields the equalities

$$\begin{tikzcd} 1_{[1]} \arrow[r, mapsto, "\alpha_0"] \arrow[d, mapsto, "d_0"'] & f \arrow[d, mapsto, "d_0"] \\ 1_{[1]} \circ d^0 = 1 \arrow[r, mapsto, "\alpha_0"'] & \alpha_0(1) = d_0(f) \end{tikzcd} \qquad \begin{tikzcd} 1_{[1]} \arrow[r, mapsto, "\beta_0"] \arrow[d, mapsto, "d_1"'] & g \arrow[d, mapsto, "d_1"] \\ 1_{[1]} \circ d^1 = 0 \arrow[r, mapsto, "\beta_0"'] & \alpha_0(0) = d_1(g) \end{tikzcd}$$

Recall that in (2), we defined $d_1(g) = \operatorname{dom} g$ and $d_0(f) = \operatorname{cod} f$. In (4), we defined $\operatorname{cod} f = y = \operatorname{dom} g$, and thus the morphisms agree. We thus have an induced map $\sigma_0 = \alpha \cup \beta : \Lambda^2_1 \to K$, which is immediately guarenteed to be defined as claimed in (4).

Now that our definitions are indeed defined, we must show $\mathcal{C}$ defines a category, as claimed.

1. First, we show the identity is indeed an identity. Let $y \in \mathcal{C}$, and let $f : x \to y$ and $g : y \to z$ be morphisms in $\mathcal{C}$. The data $(f, 1_y)$ consitutes a horn $x \xrightarrow{f} y \xrightarrow{1} y$, which fills to a 2-simplex $\tau \in K_2$ and a morphism $\sigma : \Delta^2 \to K$ given by $1_{[2]} \mapsto \tau$, such that τ witnesses $1_y \circ f$ as the composite of 1_y and f, as described in (4). We show $1_y \circ f = f$. Recall $1_y \circ f = \sigma(d^1)$. We have
$$d_0(\tau) = d_0\sigma(1_{[2]}) = \sigma d_0(1_{[2]}) = \sigma(d^0) = 1_y = s_0(y),$$
$$d_1(\tau) = d_1\sigma(1_{[2]}) = \sigma d_1(1_{[2]}) = \sigma(d^1) = 1_y \circ f,$$
$$d_2(\tau) = d_2\sigma(1_{[2]}) = \sigma d_2(1_{[2]}) = \sigma(d^2) = f.$$
Recall that there is a simplicial identity $s_i d_i = s_i d_{i+1} = 1$. We have
$$s_1(f) = s_1 d_2(\tau) = \tau,$$
and thus τ is degenerate. We also have
$$1_y \circ f = d_1(\tau) = d_1 s_1(f) = f$$
by the same simplicial identity. The converse direction is the same argument, except we find that $s_0(g) = \tau$.

2. Now we show associativity. Let
$$w \xrightarrow{f} x \xrightarrow{g} y \xrightarrow{h} z$$
be a composable triple in $\mathcal{C}$. We want to show $h(gf) = (hg)f$. Define the following 2-simplices of K to be the images of $1_{[2]}$ under the relevant fillers:
$$\sigma_{gf} = \begin{cases} d_0(\sigma_{gf}) = g, \\ d_1(\sigma_{gf}) = gf, \\ d_2(\sigma_{gf}) = f \end{cases} \qquad \sigma_{hg} = \begin{cases} d_0(\sigma_{hg}) = h, \\ d_1(\sigma_{hg}) = hg, \\ d_2(\sigma_{hg}) = g \end{cases}$$

$$\sigma_{h(gf)} = \begin{cases} d_0(\sigma_{h(gf)}) = h, \\ d_1(\sigma_{h(gf)}) = h(gf), \\ d_2(\sigma_{h(gf)}) = gf \end{cases} \qquad \sigma_{(hg)f} = \begin{cases} d_0(\sigma_{(hg)f}) = hg, \\ d_1(\sigma_{(hg)f}) = (hg)f, \\ d_2(\sigma_{(hg)f}) = f \end{cases}$$

By the same logic as earlier (and indeed, we will show this holds generally later with an inductive argument), there is a horn $\tau_0 : \Lambda^3_2 \to K$ and a filler τ defining the 3-simplex $\lambda = \tau(1_{[3]}) \in K_3$. This simplex has faces

$$d_0(\lambda) = d_0\tau(1) = \tau(d^0) = \sigma_{hg},$$

$$d_1(\lambda) = d_1\tau(1) = \tau(d^1) = \sigma_{h(gf)},$$

$$d_3(\lambda) = d_3\tau(1) = \tau(d^3) = \sigma_{gf}.$$

We first want to show $d_2(\lambda) = \sigma_{(hg)f}$. For now, we note that

$$d_0d_2(\lambda) = d_1d_0(\lambda) = d_1(\sigma_{hg}) = hg,$$

$$d_2d_2(\lambda) = d_2d_3(\lambda) = d_2(\sigma_{gf}) = f,$$

by the simplicial identity $d_id_j = d_{j-1}d_i$ for $i < j$. This defines a horn, which admits a filler. Since $\sigma_{(hg)f}$ agrees with the horn but is a 2-simplex, it necessarily is a filler, and by (2) this filler must be unique. Thus, $d_2(\lambda) = \sigma_{(hg)f}$ is the unique filler. Finally, note that by the same simplicial identity,

$$(hg)f = d_1d_2(\lambda) = d_1d_1(\lambda) = h(gf).$$

Therefore, $\mathcal{C}$ is a category, as claimed.

Finally, we show the isomorphism $K \cong N(\mathcal{C})$. By construction, we have a map $\varphi : K \to N(\mathcal{C})$ given by mapping n-simplices to the n-fold composite they witness. Of course, this is a morphism of simplicial sets basically by construction, which is checked by a simple (but annoying to type up) proof. We simply show φ is an isomorphism. It suffices to show it is an isomorphism levelwise, meaning $\varphi_n : K_n \cong N(\mathcal{C})_n$. By the Yoneda lemma, this is equivalent to showing isomorphism

$$\mathrm{s}\mathcal{S}\mathrm{et}(\Delta^n, K) \cong \mathrm{s}\mathcal{S}\mathrm{et}(\Delta^n, N(\mathcal{C}))$$

for all $n \geq 0$.

We induct on this statement. This is obvious by construction for $n = 0$ and $n = 1$, since $N(\mathcal{C})_i = K_i$ when $i = 0, 1$, so assume the inductive hypothesis for $n \geq 2$, and choose $0 < i < n$ an integer. Precomposition by the canonical inclusion $\Lambda^n_i \hookrightarrow \Delta^n$ and postcomposition by φ induce a commutative diagram

$$\begin{tikzcd} \mathrm{s}\mathcal{S}\mathrm{et}(\Delta^n, K) \arrow[r] \arrow[d] & \mathrm{s}\mathcal{S}\mathrm{et}(\Delta^n, N(\mathcal{C})) \arrow[d] \\ \mathrm{s}\mathcal{S}\mathrm{et}(\Lambda^n_i, K) \arrow[r] & \mathrm{s}\mathcal{S}\mathrm{et}(\Lambda^n_i, N(\mathcal{C})) \end{tikzcd}$$

Recall by our first part of the proof that $N(\mathcal{C})$ satisfies property (2). Thus, since K and $N(\mathcal{C})$ satisfy (2), the vertical maps are bijective. It suffices to thus show the bottom map is bijective since isomorphisms satisfy 2-out-of-3. By the inductive hypothesis, there is an isomorphism $\mathrm{s}\mathcal{S}\mathrm{et}(\Delta^{n-1}, K) \cong \mathrm{s}\mathcal{S}\mathrm{et}(\Delta^{n-1}, N(\mathcal{C}))$. Recall from earlier that any face $\partial_j\Delta^n$ is isomorphic to Δ^{n-1}, and that the horn decomposes as maps for faces. Since the faces are isomorphic to Δ^{n-1} for which the statement follows, the statement must follow for each face, and by universality the statement follows for the horn, completing the proof. ■

Remark 1.1.2.6. We may refer to n-simplices of a simplicial set K as morphisms $\sigma : \Delta^n \to K$. We recall by our use of Yoneda in the above proof that such a morphism amounts to specifying an n-simplex $\sigma \in K_n$ and an image $1_{[2]} \mapsto \sigma$, justifying this terminology.

Remark 1.1.2.7. Of course, proposition 1.1.2.5 (2) is similar to definition 1.1.2.1, but we require it only for inner horns, and we require the filler be unique. It is fairly easy to see that it would be unreasonable to assume outer horns have unique fillers in proposition 1.1.2.5. For example, let $n = 2$. A horn $\Lambda^2_2 \to N(\mathcal{C})$ is a diagram of solid arrows

$$\begin{tikzcd} & y \arrow[dr] & \\ x \arrow[ur, dotted] \arrow[rr] & & z \end{tikzcd}$$

and a filler is a dotted arrow as indicated, rendering the diagram commutative. This is of course unreasonable to ask for, but it is reasonable for *groupoids*.

Simplicial sets are thus good models for ∞-groupoids when K is a Kan complex, and are good models for regular categories when K satisfies proposition 1.1.2.5 (2). We earlier noted that ∞-groupoids and regular 1-categories are both special cases of ∞-categories, so we would expect some more general class of simplicial sets to give us a model. To see how this arises, we use our observations from proposition 1.1.2.5 and remark 1.1.2.3 to consider how a simplicial set K should be interpreted to yield an ∞-category.

First, clearly objects are vertices, morphisms are edges, and identity, domain, and codomain are the same as remark 1.1.2.3. A 2-simplex $\sigma : \Delta^2 \to K$ is thought of as a diagram

$$\begin{tikzcd} & y \arrow[dr, "\psi"] & \\ x \arrow[ur, "\phi"] \arrow[rr, "\theta"'] & & z \end{tikzcd}$$

together with a homotopy $\theta \simeq \psi\phi$ which witnesses the "commutativity" of the diagram (in higher category, commutativity of a diagram implies there are homotopies present as required to witness the composites). Simplices of higher dimension can be thought of similarly as verifying the commutivity of higher dimensional diagrams between higher level morphisms, which we will explore later as an aside from the main text.

However, this discussion has a massive issue: there is in general no way to compose two adjacent morphisms. However, inspecting the proof of proposition 1.1.2.5 gives us a property which makes this possible. A composable pair (g, f) determines a horn $\Lambda^2_1 \to K$, and the existence of their composite determines a filler $\sigma : \Delta^2 \to K$. We simply require that fillers for *inner* horns always exist.

The next problem is that of uniqueness. It is in fact fairly unnatural to require that fillers be unique. For example, recall the definition of composites of paths in a topological space, where we compress their domain along $I/2$ instead of I. We could just as easily compress one over $I/3$ and one over $2I/3$, and since the two choices are homotopic, it doesn't matter which we choose. Since they are homotopic, they should both be valid choices for the composite of these homotopies, but each corresponds to a different filler for the relevant horn. With all of these considerations, we propose the following definition.

Definition 1.1.2.8 (Quasi-category)**.** An ∞-category (in particular, a quasi-category) is a simplicial set K which admits fillers for all inner horns. More precisely, for any $0 < i < n$, for all diagrams of solid arrows

$$\begin{tikzcd}
\Lambda^n_i \arrow[r] \arrow[d, hook] & K \\
\Delta^n \arrow[ur, dotted] &
\end{tikzcd}$$

there exists at least one (not necessarily unique) dotted arrow as indicated, rendering the diagram commutative.

This definition was first proposed by Boardman and Vogt in **??**, motivated by proposition 1.1.2.5. They called them *weak Kan complexes*, and later André Joyal called them *quasi-categories*. Lurie calls them ∞-categories, although Joyal's terminology is more commonly used when referring to the specific underlying model.

Example 1.1.2.9. Any Kan complex as defined in definition 1.1.2.1 is an ∞-category. In particular, in view of remark 1.1.2.2, the singular complex $\operatorname{Sing} X$ of any topological space X is an ∞-category.

Example 1.1.2.10. By proposition 1.1.2.5, the nerve of any category is an ∞-category. We will oftentimes abuse terminology and regard 1-categories $\mathcal{C}$ as ∞-categories by identifying them with their nerve $N(\mathcal{C})$. By way of this identification, we may view 1-category theory as a special case of the theory of ∞-categories.

Notation 1.1.2.11. We will use two different notations for simplicial sets. When viewing a simplicial set K as an ∞-category, we will generally denote it with a calligraphic letter ($\mathcal{C}, \mathcal{D}, \mathcal{E} \ldots$), and otherwise we denote them by capital roman letters ($K, X, Y \ldots$). We may also employ the latter notation when comparing different models for ∞-categories, to avoid confusion.

Before we continue with the text, I want to make a couple remarks that will be useful for later.

Proposition 1.1.2.12. *Let K be a simplicial set. Specifying a horn $\Lambda^n_i \to K$ amounts to specifying $(n-1)$-simplices τ_j for all $0 \leq j \neq i \leq n$ which satisfy the compatibiltiy condition $d_k(\tau_j) = d_{j-1}(\tau_k)$ for all $k < j$. In particular, the horn is defined by $d^j \mapsto \tau_j$.*

Proof. Recall that unions are pushouts in $\mathcal{S}\mathrm{Set}$, meaning morphisms $\Lambda^n_i \to K$ are precisely pushout diagrams

$$\begin{tikzcd}
\bigcap_{j\in[n], j\neq i} \partial_j \Delta^n \arrow[r, hook] \arrow[d, hook] & \partial_j \Delta^n \arrow[d, hook] \arrow[ddr, bend left] & \\
\partial_k \Delta^n \arrow[r, hook] \arrow[drr, bend right] & \bigcup_{j\in[n], j\neq i} \partial_j \Delta^n = \Lambda^n_i \arrow[dr, dashed] & \\
& & K
\end{tikzcd}$$

where the pushout is over all $j \in [n] \setminus \{i\}$, but it is hard to draw this many arrows so we only draw two. The case for $n = 1$ is trivial, so we assume $n \geq 2$.

Recall from the proof of proposition 1.1.2.5 that there is an isomorphism of simplicial sets $\partial_j \Delta^n \cong \Delta^{n-1}$ by mapping $\delta^j \alpha \mapsto \alpha$, which is an isomorphism by the epi-mono factorization in $\mathbf{\Delta}$ ([ML98, Tal25]). Note that the map $\delta^j \mapsto \tau_j$ decomposes as

$$\begin{tikzcd} \partial_j \Delta^n \arrow[r, "\cong", "d^j\alpha \mapsto \alpha"'] \arrow[rr, bend left, "d^j \mapsto \tau_j"] & \Delta^{n-1} \arrow[r, "1_{[n-1]} \mapsto \tau_j"'] & K, \end{tikzcd}$$

where we recall that $1_{[n-1]} \mapsto \tau_j$ is a map of simplicial sets by the Yoneda lemma, as noted in the proof of proposition 1.1.2.5. Thus, $d^j \mapsto \tau_j$ decomposes as a morphism of simplicial sets, and thus is a morphism of simplicial sets for each j. Thus by universality of pushout, and by levelwise computation of universals in functor categories ([ML98]), the claimed morphism $\sigma : \Lambda_i^n \to K$ exists precisely when the above pushout diagram commutes.

First, we prove the easy direction. Let $k < j$, and suppose the diagram commutes, meaning there is indeed a morphism $\sigma : \Lambda_i^n \to K$ with $d^j \mapsto \tau_j$ for each $j \neq i$ in $[n]$. Since Λ_i^n is a simplicial subset of Δ^n, faces are given by precomposition. Since σ is a morphism of simplicial sets, we have the string of equalities

$$d_k(\tau_j) = d_k\sigma(d^j) = \sigma d_k(d^j) = \sigma(d^j d^k) \overset{!}{=} \sigma(d^k d^{j-1}) = \sigma d_{j-1}(d^k) = d_{j-1}\sigma(d^k) = d_{j-1}(\tau_k),$$

where the equality ! is given by the cosimplicial identity $d^j d^k = d^k d^{j-1}$ for $k < j$.

Now we prove the converse. Suppose the compatibility condition holds. We do this in three steps:

1. We show for $n \geq 2$, the intersection $\bigcap_{j \neq i} \partial_j \Delta^n$ has precisely one 0-simplex i, and all higher levels m have precisely one m-simplex $s_m s_{m-1} \ldots s_0(i)$.
2. We show the maps $\partial_{j,k}\Delta^n \to K$ agree on i.
3. We show this implies the maps $\partial_{j,k}\Delta^n \to K$ agree on the full intersection.

(1) First, we show i is the unique 0-simplex. Note that the 0-simplices of $\Delta^{n-1} = [n-1]$, and the 0-simplices of each $\partial_j \Delta^n$ are thus $d^j[n-1]$. Since $j \neq i$, none of the cofaces d^j skip i, meaning $i \in \operatorname{im} d^j$ for all j, and thus i is a 0-simplex. Uniqueness follows from the fact that if $k \in [n-1]$ and $k \neq i$, then there exists $\partial_k \Delta^n$ in the union, the 0-simplices of which are $d^k[n-1]$, which excludes k by definition.

Now we show all higher m-simplices are the degenerate simplices $s_m \ldots s_0(i)$ by induction. A 1-simplex σ must have its faces $d_0(\sigma), d_1(\sigma)$ be 0-simplices of the union, meaning both its faces must be i. Recall that since we are working in a simplicial subset of Δ^n, faces are precomposition with cofaces, so this implies

$$0 \mapsto 0 \mapsto \sigma(0) = 0 \mapsto 1 \mapsto \sigma(1) = i,$$

and thus σ is necessarily the map $0, 1 \mapsto i$. Thus there is a unique 1-simplex. This can be viewed as the degenerate simplex

$$s_0(i) = 0, 1 \mapsto 0 \mapsto i,$$

giving the required form. Now assume the inductive hypothesis for m. Since $s_m \ldots s_0(i)$ is the unique m-simplex, all faces of an $(m+1)$-simplex σ must be $s_m \ldots s_0(i)$. By the same logic as above, we immediately see that σ is the map $[m+1] \mapsto i$, and thus admits the representation $s_{m+1} \dots s_0(i)$, concluding this section of the proof.

(2) Let $\sigma_j : \partial_j \Delta^n \to K$ and $\sigma_k : \partial_k \Delta^n \to K$ be the maps in question. Since n is finite, it suffices to prove the statement for the case $k = j+1$, along with the additional case $k = i+1$ and $j = i-1$.

1. ($i < j$) Since $i < j$, precomposing by d_j and d_{j+1} is the identity on $i : [0] \to [n-1]$, so undoing this precomposition gives us i as an $(n-1)$ simplex in Δ^{n-1}. In particular, it suffices to consider the maps $\sigma'_j : \Delta^{n-1} \to K$ and $\sigma'_{j+1} : \Delta^{n-1} \to K$ given by inverting the isomorphism $\partial_j \Delta^n \cong \Delta^{n-1}$. The ith vertex of the simplices $\tau_j = \sigma'_j(1_{n-1})$ and $\sigma'_{j+1}(1_{n-1})$ are the respective images of i in K. The compatibility condition gives us $d_j(\tau_j) = d_j(\tau_{j+1})$. Note that
$$d_j(\tau_j) = d_j \sigma'_j(1) = \sigma'_j(d^j).$$
Since $i < j$, recall that precomposing by d^j is an identity on the ith vertex, and by the same logic since $i < j+1$, we must have that the jth face preserves the ith vertex of τ_{j+1}. However, since $d_j(\tau_j) = d_j(\tau_{j+1})$ is a strict equality, they must have the *same* ith vertex. Thus, they agree on i,

2. ($j < i-1$) The logic for this part is the same as the above part, except d_j and d_{j+1} are no longer the identity on i. Since $j, j+1 < i-1$, they both act the same way on it, mapping $i-1$ to i. The proof is concluded from this point in the same way as above.

3. ($j = i-1$) We must now show σ_{i-1} and σ_{i+1} agree on i. Since $i-1 < i < i+1$, we have that d_{i-1} maps $i-1 \mapsto i$, and d_{i+1} maps $i \mapsto i$. Thus, we must show the $(i-1)$th vertex of τ_{i-1} is the ith vertex of τ_{i+1}, by the same logic as above. The coherence condition gives us that $d_{i-1}(\tau_{i+1}) = d_i(\tau_{i-1})$. Since $i-1 < i$, the $(i-1)$th vertex of τ_{i-1} is preserved. Since $i-1 < i+1$, the ith vertex of τ_{i+1} corresponds to the $(i-1)$th vertex of $d_{i-1}(\tau_{i+1})$, since we drop the original $(i-1)$th vertex, and the other vertices shift down. The strict equality concludes the proof.

(3) Keeping notation as before, we clearly have by inducting on the morphism condition for simplicial sets that for each m-simplex α in the intersection,
$$\sigma_j(\alpha) = \sigma_j s_m \dots s_0(i)) = s_m \dots s_0 \sigma_j(i) = s_m \dots s_0 \sigma_k(i) = \sigma_k s_m \dots s_0(i) = \sigma_k(\alpha).$$
■

Corollary 1.1.2.13. *Let K be a simplicial set. A filler for a horn $\sigma_0 : \Lambda^n_i \to K$ is an n-simplex $\tau \in K_n$ such that the morphism of simplicial sets $\sigma : 1_{[n]} \mapsto \tau$ agrees with the horn Λ^n_i in the sense that for each $j \neq i$ and each face $d^j : [n-1] \to [n]$,*
$$\sigma_0(d^j) = \sigma(d^j).$$

Proof. By the Yoneda lemma, morphisms $\sigma : \Delta^n \to K$ correspond to a choice of an n-simplex $\tau \in K_n$ and a map $1_{[n]} \mapsto \tau$. The restriction $\sigma|\Lambda^n_i$ gives a horn in K, and by proposition 1.1.2.12 this amounts to a choice τ_j for each $j \neq i$ satisfying the compatibiltiy condition, and mapping $d^j \mapsto \tau_j$. By proposition 1.1.2.12, this horn is thus the given horn σ_0 precisely when $\sigma|\Lambda^n_i$ and σ_0 agree on faces, and thus σ and σ_0 agree on $(\neq i)$ faces. ■

1.1.3 Equivalences of Topological Categories

We return for a short time to our simpler notion of topological categories, in order to get a feel for what an ∞-categorical notion of an *equivalence of categories* should be. Not all Kan complexes are isomorphic to singular sets, and similarly not all CW complexes are homeomorphic to geometric realizations of simplicial sets. However, these statements *are all true* if instead of working with isomorphisms, we work with homotopy equivalences. We want to define a *functor* $F : \mathcal{C} \to \mathcal{D}$ which can be viewed as an equivalence in the same manner. Of course, the naive way to do this is to require that F induce isomorphisms of mapping spaces, meaning it is an equivalence of topological categories.

Definition 1.1.3.1. A functor $F : \mathcal{C} \to \mathcal{D}$ between topological categories is a *strong equivalence* if it is an equivalence in the sense of enriched category theory, meaning the induced continuous maps $F : \mathrm{Map}_{\mathcal{C}}(x, y) \to \mathrm{Map}_{\mathcal{D}}(F(x), F(y))$ are isomorphisms, and each $d \in \mathcal{D}$ is isomorphic to some $F(c)$ with $c \in \mathcal{C}$.

In classical category theory, this would be a perfectly fine definition, but for our cases it is still too strong. In particular, the induced morphisms between topological spaces are *isomorphisms*, not homotopy equivalences, which we would expect them to be. The obvious next idea is to recall some model category theory. Quillen equivalent model categories are generally considered to have the same homotopy theory, but the Quillen functors need not actually be equivalences; they need only lower to equivalences of the homotopy categories. Since geometric realization and the singular complex form a Quillen equivalence bewteen $\mathrm{s}\mathcal{S}\mathrm{et}$ and $\mathcal{T}\mathrm{op}$, this line of thought is very promising. However, this requires a definition of a homotopy category. To this end, we propose definition 1.1.3.2, which seems quite reasonable given that we are working with *weak* homotopy equivalences. Recall that $\pi_0 : \mathcal{T}\mathrm{op} \to \mathcal{S}\mathrm{et}$ maps topological spaces to their set of path components.

Definition 1.1.3.2. Let $\mathcal{C}$ be a topological category. The *homotopy category* $\mathrm{h}\mathcal{C}$ of $\mathcal{C}$ is defined by

- $\mathrm{Obj}(\mathrm{h}\mathcal{C}) = \mathrm{Obj}(\mathcal{C})$;
- $\mathrm{h}\mathcal{C}(x, y) = \pi_0 \,\mathrm{Map}_{\mathcal{C}}(x, y)$;

and composition is inherited by functoriality of π_0. In particular, $\mathrm{h}\mathcal{C} = \mathrm{im}\,\pi_0$.

Example 1.1.3.3. Let $\mathcal{C}$ be the category of CW complexes, with mapping spaces given by the k-ification of the compact-open topology. The homotopy category thereof is quite important, is denoted by $\mathcal{H}$, and is called the *homotopy category of spaces*.

Using $\mathcal{H}$, we can rewrite definition 1.1.3.2 in a useful way. Recall:

Definition 1.1.3.4. A map $f : X \to Y$ of topological spaces is a *weak homotopy equivalence* if it induces a bijection $\pi_0(f) : \pi_0(X) \to \pi_0(Y)$ on path components, and isomorphisms on all homotopy groups

$$\pi_i(f) : \pi_i(X, x) \to \pi_i(Y, f(x)).$$

Classical homotopy theory tells us for each $X \in \mathcal{CG}$, there is a CW complex X' and a weak homotopy equivalence $\varphi : X' \to X$. This space X' is well-defined up to canonical weak homotopy equivalence, meaning there is a functor $\theta : \mathcal{CG} \to \mathcal{H}$ given by $X \mapsto [X] \ni X'$. This is precisely the localization functor, meaning it is universal with respect to inverting weak homotopy equivalences. Recall that lax monoidal functors $F : \mathcal{C} \to \mathcal{C}'$ induce 2-functors $F : \mathcal{C}\text{-}\mathcal{C}\mathrm{at} \to \mathcal{C}'\text{-}\mathcal{C}\mathrm{at}$, and since θ fairly clearly preserves products, θ induces a 2-functor $\mathcal{C}\mathrm{at}_{\mathrm{top}} \to \mathcal{H}\text{-}\mathcal{C}\mathrm{at}$. In particular, topological categories $\mathcal{C}$ are mapped to $\mathcal{H}$-enriched categories. It seems reasonable, since θ is localization, to define the image of $\mathcal{C}$ as the *homotopy category* $\mathrm{h}\mathcal{C}$ of $\mathcal{C}$. Recalling how the functor $\mathcal{C}\mathrm{at}_{\mathrm{top}} \to \mathcal{H}\text{-}\mathcal{C}\mathrm{at}$ works, this gives

1. $\mathrm{Obj}(\mathrm{h}\mathcal{C}) = \mathrm{Obj}(\mathcal{C})$;

2. For $x, y \in \mathcal{C}$, we have
$$\mathrm{Map}_{\mathrm{h}\mathcal{C}}(x, y) = [\mathrm{Map}_{\mathcal{C}}(x, y)],$$
where we continue to use the equivalence class $[-]$ notation to denote a choice of a weakly homotopic equivalent CW complex.

3. Composition is given by functoriality of $\theta : \mathcal{CG} \to \mathcal{H}$.

Note here that θ maps morphisms to their (regular) homotopy classes. For a refresher on enriched category theory, we recommed chapter 3 of [Rie14], but for a more general account of the theory see [Kel05].

Intuition 1.1.3.5. Departing slightly from the book here to improve my intuition. θ is simply localization at the class of weak homotopy equivalences, which is the class of weak equivalences in $\mathcal{T}\mathrm{op}$. By Whitehead's theorem, we may choose $\mathcal{H}$ to have CW complexes as objects up to equivalence of categories. Of course, between cofibrant and fibrant objects, this agrees with the regular notion of a homotopy equivalence. In particular, each $\pi_n(X, x)$ identifies homotopic maps, making homotopy equivalence a stronger condition than weak homotopy equivalence.

Remark 1.1.3.6. The equivalence of our definition of $\mathrm{h}\mathcal{C}$ by θ and by definition 1.1.3.2 follows by noting that there is a canonical isomorphism $\pi_0(X) \cong \mathrm{Map}_{\mathcal{H}}(*, [X])$, recovering $\mathrm{h}\mathcal{C}$ as the underlying category of the $\mathcal{H}$-enriched version.

Taking our cue from model category theory, we now think to define an equivalence as a functor that descends to an equivalence of categories $\mathrm{h}\mathcal{C} \simeq \mathrm{h}\mathcal{D}$.

Definition 1.1.3.7. Let $F : \mathcal{C} \to \mathcal{D}$ be a functor between topological categories. We say F is a *weak equivalence* if the induced functor $\mathrm{h}F : \mathrm{h}\mathcal{C} \to \mathrm{h}\mathcal{D}$ is an equivalence of $\mathcal{H}$-categories. Equivalently, F is an equivalence if and only if

1. for each $x, y \in \mathcal{C}$, the map $\mathrm{Map}_{\mathcal{C}}(x, y) \to \mathrm{Map}_{\mathcal{D}}(F(x), F(y))$ is a weak homotopy equivalence;
2. Each $d \in \mathcal{D}$ is isomorphic to $F(c)$ for some $c \in \mathcal{C}$.

Here we recall that an equivalence of $\mathcal{H}$-categories must induce isomorphisms on hom-objects, and since θ inverts weak homotopy equivalences, the maps can equivalently be weak homotopy equivalences in $\mathcal{CG}$. Weak equivalences are the correct notion of equivalences in the context of higher category theory. We will thus refer to them simply as *equivalences* as we continue. Of course, two topological categories are *equivalent* if an equivalence between them exists. However, before we continue, note that the homotopy category of a topological category $\mathcal{C}$ does *not* determine $\mathcal{C}$ up to equivalence.

1.1.4 Simplicial Categories

Now that we have a good notion of equivalences of topological categories, we want to show the theories of quasi-categories and topological categories are equivalent. It will be extremely beneficial to introduce the theory of *simplicial categories*, which is not only related to both approaches, but will also motivate definitions similar to those of section 1.1.3 for quasi-categories.

Definition 1.1.4.1. A *simplicial category* is a $\mathrm{s}\mathcal{S}\mathrm{et}$-category; that is, it is a category enriched over the category of simplicial sets. We denote the category of simplicial categories by $\mathrm{s}\mathcal{C}\mathrm{at}$.

Remark 1.1.4.2. Every simplicial category $\mathcal{C}$ forms a simplicial object $\mathcal{C}_\bullet$ in $\mathcal{C}\mathrm{at}$ by taking $\mathrm{Obj}(\mathcal{C}_n) = \mathrm{Obj}(\mathcal{C})$, and $\mathcal{C}_n(x, y) = \mathcal{C}(x, y)_n$. Faces and degeneracies are defined by the action of the faces and degeneracies of $\mathcal{C}$ within its hom-objects. Conversely, simplicial objects $\mathcal{C}_\bullet$ in $\mathcal{C}\mathrm{at}$ give rise to simpliical categories $\mathcal{C}$ *precisely when* $\mathrm{Obj}(\mathcal{C}_n) = \mathrm{Obj}(\mathcal{C}_m)$ for all n, m. We say this means the *underlying simplicial set* of objects is constant. Also note that every category is a simplicial category by considering the discrete simplicial set on each hom-set.

Of course, the Quillen adjunction $|-| \dashv \mathrm{Sing} : \mathrm{s}\mathcal{S}\mathrm{et} \to \mathcal{C}\mathcal{G}$ makes the relationship between topological and simplicial categories simple to describe. Both of these functors preserve finite products, and thus are lax monoidal functors. Thus, given a simplicial category $\mathcal{C}$, there is a topological category $|\mathcal{C}|$ given by

- $\mathrm{Obj}(\mathcal{C}) = \mathrm{Obj}(|\mathcal{C}|)$;
- For all $x, y \in \mathcal{C}$, $\mathrm{Map}_{|\mathcal{C}|}(x, y) = |\mathrm{Map}_{\mathcal{C}}(x, y)|$.

We also obtain for each topological category $\mathcal{C}$ a simplicial category $\mathrm{Sing}\,\mathcal{C}$ given in the same way by

- $\mathrm{Obj}(\mathcal{C}) = \mathrm{Obj}(\mathrm{Sing}\,\mathcal{C})$;
- For all $x, y \in \mathcal{C}$, $\mathrm{Map}_{\mathrm{Sing}\,\mathcal{C}}(x, y) = \mathrm{Sing}(\mathrm{Map}_{\mathcal{C}}(x, y))$.

In particular, $|-| \dashv \mathrm{Sing} : \mathrm{s}\mathcal{S}\mathrm{et} \to \mathcal{C}\mathcal{G}$ raises to an adjunction $|-| \dashv \mathrm{Sing} : \mathrm{s}\mathcal{C}\mathrm{at} \to \mathcal{C}\mathrm{at}_{\mathrm{top}}$. To show the theories of topological and simplicial categories are equivalent, since we have modeled them homotopy-theoretically, we once again turn to homotopy theory. We previously defined weak equivalences of topological categories, so now we would like to define them between simplicial categories. We recall that weak equivalences between simplicial sets are maps $f : X \to Y$ which induce weak homotopy equivalences $|f| : |X| \to |Y|$. Inverting these gives a homotopy category which we also denote $\mathcal{H}$, since the Quillen equivalence $|-| \dashv \mathrm{Sing}$ defines an equivalence of categories between the homotopy categories for CW complexes and for simplicial sets.

We can once again define $\mathrm{h}\mathcal{C}$ for simplicial categories by applying the functor $\mathrm{s}\mathcal{S}\mathrm{et} \to \mathcal{H}$ and considering the induced $\mathcal{H}$-category. There are canonical isomorphisms $\mathrm{h}\mathcal{C} \cong \mathrm{h}\,|\mathcal{C}|$ and $\mathrm{h}\mathcal{D} \cong \mathrm{h}\,\mathrm{Sing}\,\mathcal{D}$, meaning that our choice of notation offers little risk of confusion. It is at this point fairly obvious that topological categories and simplicial categories are equivalent, but we may finalize this fact by introducing the following definition.

Definition 1.1.4.3. A functor $F : \mathcal{C} \to \mathcal{D}$ between simplicial categories is an *equivalence* if $\mathrm{h}F : \mathrm{h}\mathcal{C} \to \mathrm{h}\mathcal{D}$ is an equivalence of $\mathcal{H}$-categories.

In particular, since $\mathrm{h}\mathcal{C} \cong \mathrm{h}\,|\mathcal{C}|$, a functor $F : \mathcal{C} \to \mathcal{D}$ is an equivalence if and only if the geometric realization $|F| : |\mathcal{C}| \to |\mathcal{D}|$ is an equivalence. Even better, we note by these remarks that in the adjunction $|-| \dashv \mathrm{Sing} : \mathrm{s}\mathcal{C}\mathrm{at} \to \mathcal{C}\mathrm{at}_{\mathrm{top}}$, the unit and counit maps

$$\mathcal{C} \to \mathrm{Sing}\,|\mathcal{C}|\,, \qquad |\mathrm{Sing}\,\mathcal{C}| \to \mathcal{C}$$

induce *isomorphisms of homotopy categories*. Not only is this exactly what we expect a Quillen equivalence to look like (without defining a whole model structure), but we also note that we may freely replace topological categories with simplicial categories up to equivalence, giving us our equivalence of models for ∞-categories.

1.1.5 Comparing ∞-Categories with Simplicial Categories

In section 1.1.5, we showed simplicial categories and topological categories were equivalent as models for ∞-categories. Showing this amounted to showing that there is an adjunction which is similar to a Quillen equivalence in that it preserves (weak) equivalences, and that the unit and counit are natural (weak) equivalences, allowing us to replace topological categories $\mathcal{C}$ with simplicial categories $\operatorname{Sing}\mathcal{C}$, and simplicial categories $\mathcal{C}$ with topological categories $|\mathcal{C}|$, up to equivalence. Our proof that simplicial categories are equivalent to quasi-categories should be similar. I also remark here that these equivalences *are indeed Quillen equivalences*, but since this first section of the book is more of an overview, we're glossing over many of the details, including (co)fibrations. Our proof will involve the *simplicial nerve functor* $N : \mathrm{s}\mathcal{C}\mathrm{at} \to \mathrm{s}\mathcal{S}\mathrm{et}$, and the *Joyal rigification functor* $\mathfrak{C} : \mathrm{s}\mathcal{S}\mathrm{et} \to \mathrm{s}\mathcal{C}\mathrm{at}$. Credit to the nLab ([nLa25]) for the name of $\mathfrak{C}$.

Recall that the nerve of a category $\mathcal{C}$ has n-simplices $N(\mathcal{C})_n$ given by chains of n composable morphisms. This is equivalently viewed as a functor $[n] \to \mathcal{C}$, since the datum of such a functor amounts to a choice of $n+1$ objects, and n composable morphisms between them. Also recalling that n-simplices are precisely morphisms $\Delta^n \to N(\mathcal{C})$, we can characterize n-simplices via the natural isomorphism

$$\mathrm{s}\mathcal{S}\mathrm{et}(\Delta^n, N(\mathcal{C})) \cong \mathcal{C}\mathrm{at}([n], \mathcal{C}) = N(\mathcal{C})_n.$$

We would like to promote this to a relationship between simplicial categories. Rather than waste time trying to figure out how to do this, we are just going to note that we end up with the wrong idea. The promotion we get does not involve the simplicial structure of $\mathcal{C}$, and instead it's more suitable to replace $[n]$ with a different category $\mathfrak{C}[\Delta^n]$.

Definition 1.1.5.1. Let J be a finite nonempty linearly ordered set. The simplicial category $\mathfrak{C}[\Delta^J]$ is defined by

1. The objects of $\mathfrak{C}[\Delta^J]$ are the elements of J.
2. For $i, j \in J$, mapping spaces are given by

$$\mathrm{Map}_{\mathfrak{C}[\Delta^J]}(i,j) = \begin{cases} \varnothing, & j < i \\ N(P_{i,j}), & i \leq j \end{cases},$$

   where P_{ij} is the poset

$$P_{ij} = \{I \subseteq J \mid i, j \in I \wedge (\forall k \in I)(i \leq k \leq j)\}$$

   ordered by inclusion. Note that elements I have $\min I = i$ and $\max I = j$.

3. If a mapping space is empty then a composition morphism to/from it is the empty map, so let $i_0 \leq \ldots \leq i_n$. Composition morphisms

$$\mathrm{Map}_{\mathfrak{C}[\Delta^j]}(i_0, i_1) \times \cdots \times \mathrm{Map}_{\mathfrak{C}[\Delta^J]}(i_{n-1}, i_n) \to \mathrm{Map}_{\mathfrak{C}[\Delta^J]}(i_0, i_n)$$

   are given by taking the nerve functor on the morphism of poset categories

$$P_{i_0,i_1} \times \cdots \times P_{i_{n-1},i_n} \to P_{i_0,i_n},$$

$$(I_1, \ldots, I_n) \mapsto \bigcup_{j \in [n]} I_j.$$

Lurie offers a short description of the topological category $|\mathfrak{C}[\Delta^n]|$ to provide some intuition. The objects of $|\mathfrak{C}[\Delta^n]|$ are the objects of $\mathfrak{C}[\Delta^n]$ are the objects of $[n]$. The space $\mathrm{Map}_{|\mathfrak{C}[\Delta^n]|}(i,j)$ is homeomorphic to a cube when $i \leq j$, and empty otherwise. It may be identified with the set of functions $p : \{k \in [n] \mid i \leq k \leq j\} \to I$ such that $p(i) = p(j) = 1$, and composition is given by concatenation of functions.

Now we try to understand the simplicial category $\mathfrak{C}[\Delta^n]$ itself, and how it relates to $[n]$. Once again, $\mathfrak{C}[\Delta^n]$ and $[n]$ have the same objects. In $[n]$, there are unique morphisms $i \to j$ for $i \leq j$, and composites are defined by virtue of this uniqueness. We denote by q_{ij} the morphism $i \to j$, and note that $q_{jk} \circ q_{ij} = q_{ik}$ for $i \leq j \leq k$.

In $\mathfrak{C}[\Delta^n]$, we have vertices $p_{ij} \in \mathrm{Map}_{\mathfrak{C}[\Delta^n]}(i,j)$ for each $i \leq j$ given by $\{i,j\} \in P_{ij}$. Of course $p_{jk} \circ p_{ij} = \{i,j,k\} \neq p_{ik}$ for $i \leq j \leq k$, unless we have a degenerate case with either $i = j$ or $j = k$.

1.2 The Language of Higher Category Theory

This section is devoted to defining many elementary 1-categorical notions in an ∞-categorical setting. Our main results are the defintions of the opposite category, mapping spaces, the homotopy category, the language of objects, morphisms, and equivalences, homotopy, (full, faithful, essentially surjective) functors, joins, slice and coslice categories, subcategories, initial and terminal objects, limits and colimits, and presentations.

1.2.1 The Opposite of an ∞-Category

Definition 1.2.1.1. Lurie's definition in [Lur09] made little sense, so I pulled from [Lur25, 003L]. Let S be a simplicial set. Define $\mathrm{Op} : \mathbf{\Delta} \to \mathbf{\Delta}$ by $[n] \mapsto [n]$ and for each $\alpha : [m] \to [n]$, we define $\mathrm{Op}(\alpha) : [m] \to [n]$ by mapping $i \mapsto n - \alpha(m-i)$. We then define $S^{\mathrm{op}} = S \circ \mathrm{Op}$.

Equivalently, and more concretely, we define $S^{\mathrm{op}}_n = S_n$, and faces and degeneracies are given by

$$(d_i : S^{\mathrm{op}}_n \to S^{\mathrm{op}}_{n-1}) = (d_{n-1} : S_n \to S_{n-1}),$$
$$(s_i : S^{\mathrm{op}}_n \to S^{\mathrm{op}}_{n+1}) = (s_{n-i} : S_n \to S_{n+1}).$$

Claim 1.2.1.2. *Let S be an ∞-category. Then so is S^{op}.*

Proof. We show for $0 < i < n$, S fills horns $\Lambda^n_i \to S$ if and only if S^{op} fills horns Λ^n_{n-i}. First, we must of course show S^{op} is a simplicial set, and since $S \circ \mathrm{Op}^{\mathrm{op}} : \mathbf{\Delta}^{\mathrm{op}} \to \mathcal{S}\mathrm{et}$, it suffices to show faces and degeneracies are indeed described as claimed in definition 1.2.1.1. For this, it suffices to show $\mathrm{Op}(d^i) = d^{n-i}$ and $\mathrm{Op}(s^i) = s^{n-i}$ in $\mathbf{\Delta}$. We prove this only for faces, since the proof for degeneracies is the same. Let $j \in [n-1]$, and note that by definition 1.2.1.1,

$$\mathrm{Op}(d^i)(j) = n - d^j(n-1-j) = \begin{cases} n-n+1+j, & n-1-j < i \\ n-n+1+j-1, & n-1-j \geq i \end{cases}.$$

Note that the top case simplifies to $j+1$ and the bottom to j. Also note that $n-1-j < i$ if and only if $n-i < j+1$ if and only if $n-i \leq j$. Similarly, $n-1-j \geq i$ if and only if $n-i \geq j+1$ if and only if $n-i > j$. This is precisely the definition of d^{n-i}. Thus, the definitions as given are in agreement, and in particular this shows Op is indeed a functor. Thus, S^{op} is a simplicial set.

We now show the horn filling condition. Let S be an ∞-category. This first involves showing $\mathrm{Op} : \mathbf{\Delta} \to \mathbf{\Delta}$ is an isomorphism. Indeed, it is it's own inverse, since

$$(\mathrm{Op} \circ \mathrm{Op})[n] = [n], \qquad (\mathrm{Op} \circ \mathrm{Op})(d^i) = \mathrm{Op}(d^{n-i}) = d^{n-(n-i)} = d^i,$$
$$(\mathrm{Op} \circ \mathrm{Op})(s^i) = \mathrm{Op}(s^{n-i}) = s^{n-(n-i)} = s^i.$$

By Yoneda, precomposition gives an isomorphism $(\mathrm{Op}^{\mathrm{op}})^* : \mathcal{C}\mathrm{at}(\boldsymbol{\Delta}^{\mathrm{op}}, -) \cong \mathcal{C}\mathrm{at}(\boldsymbol{\Delta}^{\mathrm{op}}, -)$. Temporarily passing to a higher universe, we may plug $\mathcal{S}\mathrm{et}$ into the blank, giving $(\mathrm{Op}^{\mathrm{op}})^* : \mathrm{s}\mathcal{S}\mathrm{et} \cong \mathrm{s}\mathcal{S}\mathrm{et}$. We have thus reduced the problem to showing for any inner horn $\sigma_0 : \Lambda^n_i \to S^{\mathrm{op}}$, the corresponding map $\sigma_0^{\mathrm{op}} : (\Lambda^n_i)^{\mathrm{op}} \to S$ is also a horn, and that $(\Delta^n)^{\mathrm{op}} \cong \Delta^{\mathrm{op}}$.

We first show $\Delta^n \cong (\Delta^n)^{\mathrm{op}}$. Actually I dont wanna prove this the map is precompositing by $i \mapsto m - i$ for $i \in [m]$ go do the proof urself ok cool

We note that restricting this isomorphism along a horn Λ^n_i gives $d^j \mapsto d^{n-j}$. In particular, $\Lambda^n_i \cong (\Lambda^n_{n-i})^{\mathrm{op}}$ (given by a simple inspection of elements). Since $0 < i$, we have $n - i < n$, and since $i < n$, we have $0 < n - i$. Thus, Λ^n_{n-i} is an inner horn, and admits a filler under σ_0^{op} since S is an ∞-category, completing the proof. ■

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